# Contribution Journal: Lab 10 Unsupervised Learning

ITAI 1371 · Submitted November 7, 2026

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**Group and members:** JosephClay (individual: Joseph Clay)

## 1. Which tasks or cells did you personally write or debug?

- **Data setup (cell 3):** Ran the given `make_blobs` code and confirmed the scatter plot showed 300 points in four clear groups.
- **Task 1, Elbow Method (cell 5):** Wrote the loop for k = 1 to 10, fit KMeans at each k, stored `inertia_`, and plotted the elbow curve. Inertia dropped steeply up to k = 4 and flattened after.
- **Task 2, K-Means (cell 7):** Fit KMeans with k = 4, pulled `labels_` and `cluster_centers_`, and plotted the clusters by color with the centers marked as red X's.
- **Task 3, PCA (cell 10):** Reduced Iris from 4 features to 2 components, printed explained variance (PC1 92.46%, PC2 5.31%, total 97.77%), built the results DataFrame, and plotted it by species.
- **Debugging:** Task 3 failed with `NameError: name 'sns' is not defined` because the notebook never imports seaborn. Fixed it by adding `import seaborn as sns`.
- **Knowledge Check:** Wrote answers to all three questions.
- **Final run:** Removed a stray empty cell, re-ran the notebook, and confirmed every code cell has output.

## 2. What did each teammate contribute?

No other group members. I completed all tasks independently.

## 3. Commits or files pushed to GitHub

- Repository: https://github.com/ClayClimate-AI/itai-1371-lab10-unsupervised-learning
- Files: `L10_JosephClay_Notebook.ipynb`, `L10_JosephClay_Notebook.pdf`, `L10_JosephClay_Reflection.pdf`, `L10_JosephClay_Contribution.pdf`

## 4. Decisions you made or influenced

- **Working file:** The downloaded notebook raised a SyntaxError because its line breaks were missing, so I worked from a copy with only the line breaks restored.
- **Scaling:** Kept the blob data unscaled, following the notebook hints. In the cell 3 plot, both features cover similar ranges.

- **Choosing k:** Picked k = 4 from my own elbow plot. It matched the four groups visible in the raw data.
- **Checking the 95% claim:** Knowledge Check question 3 says two components explain "over 95%." I checked that against my own output: 97.77%.
- **Repository contents:** Pushed only the deliverables. Support files and the local Python environment are excluded.

### 5. Time spent and how the work was split

Completed in one working session on November 7, 2026. All work was done by me.

**Tools:** I ran the notebook locally in VS Code using a Python virtual environment. I used Claude Code (an AI assistant) to set up the environment, explain errors, check my outputs, format my written answers, and export the PDFs. I wrote the notebook code and the answers myself.